1a

Intelligence is the capacity of learn, understand and think.

Types of Intelligence.

a) Natural Intelligence
The intelligence comes from biologically. It is evaluate from biologically. Eg: Human has capability to react when fire hot us.

b) Social Intelligence
The intelligence in a society. It awares a person how to behave in a mass of group or society.

c) Spatial Intelligence
The intelligence of analysis the visual diagrams. You can identify the things by seeing a visual.

d) Creative Intelligence
It creates creativity. It derives innovation in technology. Eg: developing a software, painting.

e) Linguistic Intelligence
It tells how to communicate eachothers. What language should be used in where.

f) Mathematical and logical Intelligence
It is the capability of solve mathematical problems and give logic about the question.

AI is the capability of machine that can think, learn and act like humans.
Importance of AI:

a) Automative System
AI automates various tasks without minimal intervention of human. For example a smart car, smart home automatically acts to its environment. That makes work fast.

b) Enhance Decision Making
AI helps to make right decision in short time. It helps in fraud detection, spam filter which directs the best option.

c) Recommendations
AI often used in entertainment, feeback system. The application Netflix, Amazon provides the movies and products as customer like and need.

d) AI can work 24/7 hours never tried as humans and increases efficiency.

e) AI can reduce the error made by humans that reduces risk in a system.

f) AI is widely used in education, survey, healthcare, banking etc.

7

1b

As AI is automative feature it is replacing a lot of automation task that used to be done by humans. It can replace some jobs but I think it cann't replace humans entirely.

- AI is a human made machine stands as Artificial Intelligence it has limited features it cann't behave as completely to humans as it has no feeling, emotions, communication skill like human.

- AI is replacing some automative jobs it may can reduce some job opportunities like in Chinese Industry Robot is used as packing purpose which displaced the job of labours. As ATM used it has displaced the banker job to provide money to consumers by withdraw. But still to conduct the machine even human effort is necessary. We have to give command to the AI.

- For example: AI helps to find difficult diseases that cann't found by doctors and help in diagnosis but even doctor presence is significant. Doctor have to analysis it and need to treat patients.

- I think it is right that some employment opportunities are missed but not whole human is displaced because human is the guider of AI to find maximum performance.

- like Chatbots, ChatGPT, Gemini, Deepseek, google assitant, herbonouli etc gives us the required answer but it is also needed a human to input question. As you give right way a input you get a right output and best workplace.

6

- I think in future human intelligence and Artificial Intelligence combine together Artificial Human Intelligence System that has common features of human and machines and produce a high quality output which peak the performance of a organization, IT company etc.
- So AI cann't displace human entirely, it acts as a assitant part of human.

2a.

Agent is the autonomous system which perceives the environment, makes decisions based on the data and perform a action to environment. Eg: Google Map

Types of Agent

a) Simple-Reflex Agent
- percept only on current data
- Eg: thermostat

b) Model Based Agent
- has memory
- eg: vaccum cleaner know where to clean

c) Goal Based Agent
- has a specific goal
- eg: google map always shows best route

d) Utility Agent
- directs to maximize performance.
- eg: choose best tools among various tools

e) Learning Agent
- learn from environment and change with time
eg: smart cars

b) Model Based Agent

- The agent which has memory.
- It analysis the previous data that are collected from environment and performs action in current.
- It acts on condition action rules like if, else, then.
- Eg: A vacuum cleaner know where to clean.

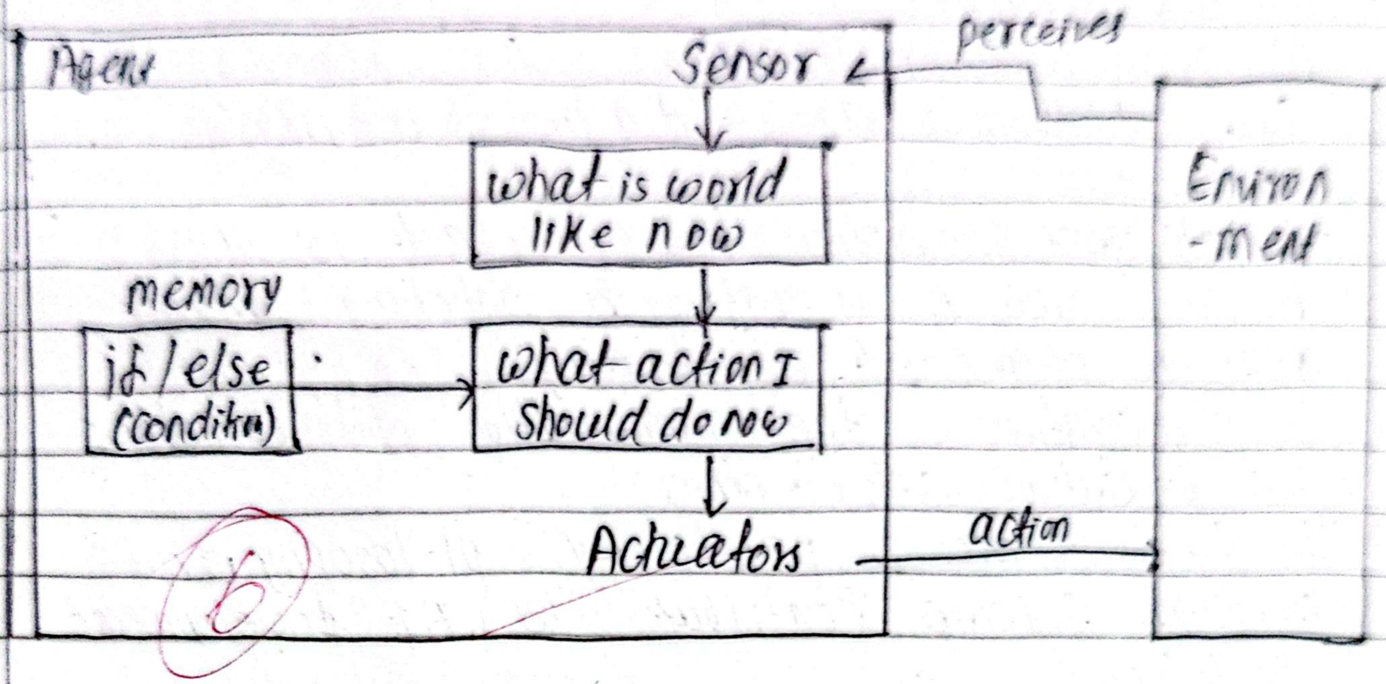

fig: Model Based Agent

- Here a sensor perceives data from it's environment.
- Then Agent analyze what world is like now. what is suitable. It see it's memory perform condition and choose what action it should perform.
- which directs Actuators and it performs action to environment.

2b

Foundations of AI means how and from where AI is founded. AI first time John Mccarthy coined the world in a seminar. It also tell that what are founded by AI.

- AI is widely used in banking for transactions, fraud detection.
- AI is used in heathcare for expert system. find hidden diseases, diagnos, successful treatments.
- Intelligence foundation: how to react to others.
- AI is used in industry as robotics which peak the output of a company.
- Social foundation: the communication with eachothers.
- Moral foundation: sets of ethics

The fields like psychology, philosophy and computer science contribute to AI development as follows:

Psychology: psychology is the field wher a psychologist can say what a person think and want. What is in his/her heart. It creats a intelligence. Thinking about others forms AI.

Philosophy:
Philosophy is the science in which a philosophist gives his/her perception. By analyzing a behaviour, environment he/she gives a concept or perception. The perception system help to develop AI.

Computer Science:
As we know computer is essential and fast growing technology in recent world. A AI is also a part of Computer Science. Development different apps, software system produce AI. for example Deepseek, Gemini, ChatGPT are the AI as a result computer science programs.

5.5

In this the three components psycology, philosophy and Computer science equally contribute to development of AI.

3a

Congitive means the ability to remember that means memory system.
Cognitive modelling is very importance part in 'Thinking Humanly' approach.

- Thinking Humanly is a approach of AI where a machines thinks like humans using logic, gives reasoning.

- As human has memory so it is possible to think logically. When there is cognitive modelling in a machine it has old memory. It can percepts to environment.

- When a machine analysis past datas, generate logic, gives reasoning, can decide what to do, it means machine can think like human. That generate 'Thinking Humanly'.

- Cognitive Modelling is very relevance and essential part for thinking humanly approach.

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• For example: A give a question to a google AI mode. It gives me a output about that. If I ask it to a new question related to the topic it gives me the answer based on this without I retype the original question. The AI automatically analyzes the past data.

Like wise Deepseek, ChatGPT also gives answer as the previously inputed data.

6.5

• It gives the platform that a machine think like human by guessing that it also related to previous output.

3b.

Well-defined problems are categorized as which is clear.

- In this we are clear about the environment.
- The solution is fixed, we know what to do, where to do and when to do.

- Well-defined problems are easier to solve. We can get fast output.

- For example: chess game, 8-puzzle game, N-queen problem etc.

Let I have a problem of 4 queen.

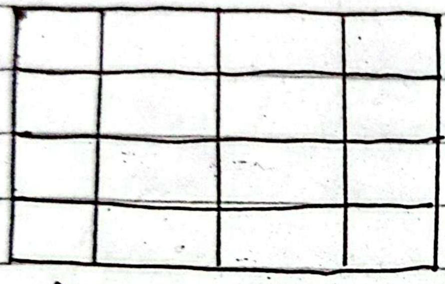

fig: 4 Queen Problem

Here I have a 4X4 box. When I need to place the 4 queens.

rules:

- every column and every row must have only one queen.
- no more two or more queens are allowed in diagonally.

Solution:

	Q		
			Q
Q			
		Q	

fig: Solution1

		Q	
Q			
			Q
	Q		

fig: Solution 2

In the above figures I have two solutions. Where:

6 - every column and row contains a Queen.
- No two queens are in diagonally.

→ Here as clear of rules we have identify the solution fastly and clearly.

→ So well defined problems are easier and fast solvable than Ill-defined problems.

→ We can get output efficiently of well-defined problems.

4a

- Turing Test is the concept given by Alan Turing.

- It is the example of 'Acting Humanly' approach of AI.

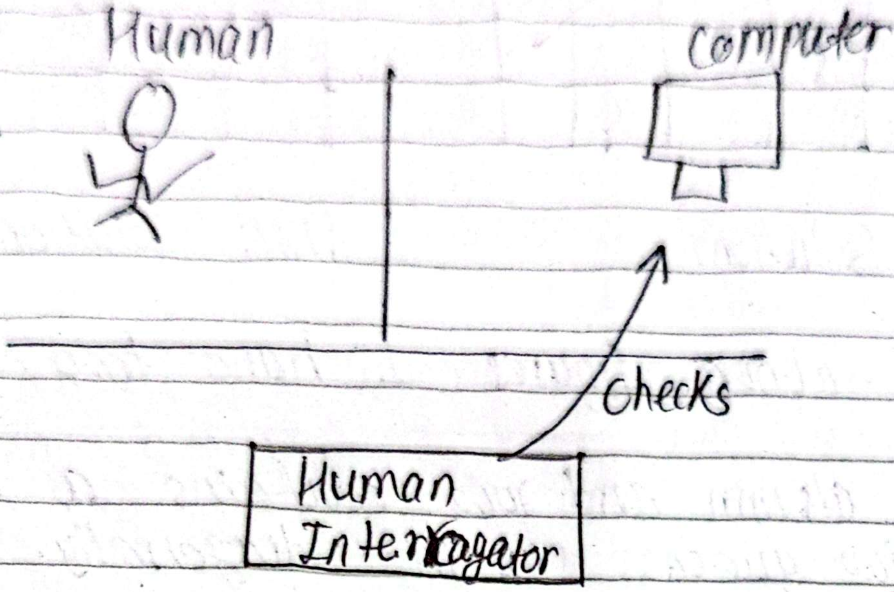

fig: Turing test

- In Turing test there are two part.
- One part has a human interogator and other has two body one is human and another is computer machine.
- The human interrogator can be person or computer.

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Additional answerbook No. 4131
Number of additional answerbooks used : .... *Invigilator's Signature*

Name of the student : .... Programme : ....
Roll No. (Class) : .... Exam : .... Grade/Semester/ .... Exam : ....
Paper : .... Date : ....

* Students are required to write answer on both sides starting from the first page of this copy.
* This answerbook must be stiched properly with the original answer book before delivering to the invigilator.

Human interrogator communicates with computer via texts, gestures etc.
- In this computer also behaves as humans.
- If the human interrogator is successful to decide that which is machine or which is humans, then turing test fails.

- If the Human Interrogator cann't decide which is human or machines and telling that the machine is human then the test is pass.

6

→ Turing test is used to find the capacity of a machine that is able or not discriminates between machine or humans.

1 b

The PEAS (Performance measure, Environment, Actuators, Sensors) are the representation of a task environment. Let taking the example of a agent like vaccum cleaner for this:

a) Performance Measure
These are the factors that ensure success of a work. It is the efficiency, reliability, effectiveness of using a agent. It shows the performance status. For example: in vaccum cleaner the performance measure are how the placed cleaned, damage of chair, furniture, remaining dirt or not etc. Perfomance can also expressed in %.

b) Environment
Environment is the surrounding or external world from which agent get data. It is the situation in which agent get contact. Example: room, wood, furniture, carpet char are environment for vaccum cleaner.

c) Actuators

They are the components which performs physical action in environment. It performs desired output Eg: dirt cleaner, ~~detector~~ wheel, suck compontent are the actuators for vaccum cleaner.

d) Sensors

Sensors are the components which senses the environment. It perceives the environment. Eg: dirt detector, cameras, file-rangers, ultrasonic waves are the example of sensors in a vaccum tube.

6.5

All the 4 components Performance measure, environment, actuators and sensor help to analyze a task of a agent.

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5a

- Learning Agents are the agents which learn from data and change over time.

- Learning agents improve over time by learn from it's previous data and decide to what act when and how that peaks performance.

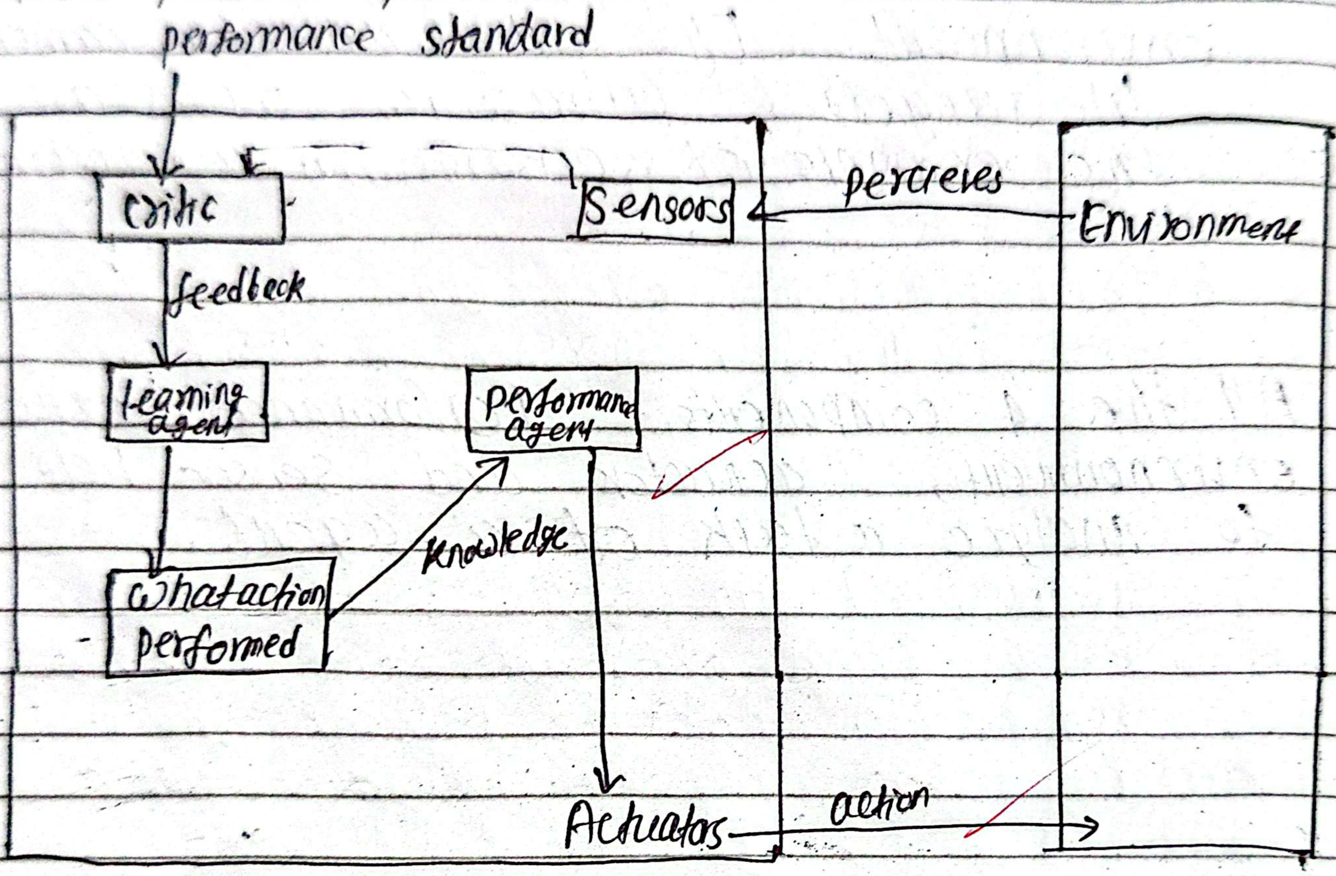


fig: Learning Agent

sensor

- Learning agent's perceives environment.
- the critic check the performance it is right or not.
- learning agent learn from data, analysises past data
- it decides what action should perform.
- permance agent perform action through actuators in environment.

- For example: A smart home system is aware about what actions is performed when. Like when person comes it opens door, when person goes it closes door. The fan itself started when a high temperature. It learne's from the data what do where and when.

6

- Learning is the best agent among all agent.

5b

Simulated annealing is a uninformed searching technique.

- It is method as which is used in metallurgy annealing.
- It is introduced to remove the limitation of Hill-climbing problem.

~~It avoids~~

- It avoids getting trapped in localmaxima as it uses worse way to travel.

- ✓ Simulated Annealing has a memory system also.

It avoids trapped in localmaxima as:

→ It first initialize a temperature.

$e^{\Delta E/T}$

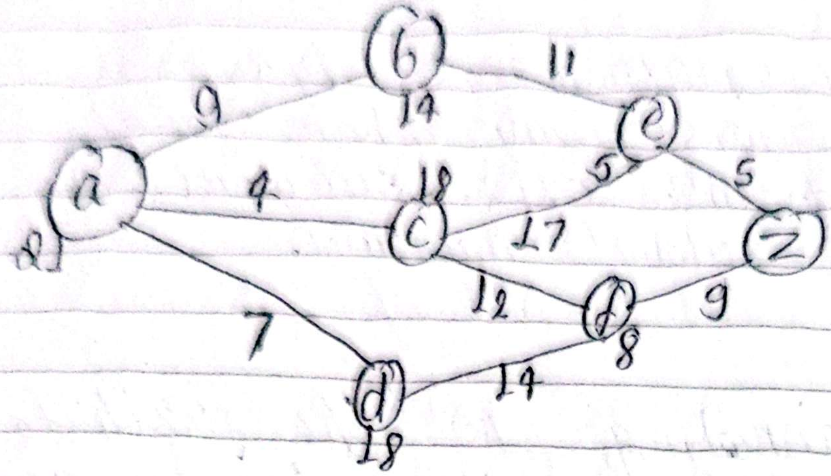
- It choose worse way of travel to avoids trapped from localmaxima because we have to reach on global maxima.
- It increases temperature until it reaches ~~to~~ $e^{\Delta E/T}$.
- After getting the peak it decreases and directs to solution.

- Simulated annealing never confused about the global maxima and local maxima.
- Here local maxima is ignored as we goto worse way where local maxima doesn't exist. Our goal is to reach to global maxima.

Simulated annealing is the path to avoid local maxima trapped and get optimal solution by reaching our destination global maxima.

6

6a A* search:



Start node: a , goal node = z

from a:

a → b , f(b) = g(b) + h(b) = 9+14 = 23 ✓
a → c , f(c) = 4 + 18 = 22 ✓
a → d , f(d) = 7+18 = 25

Choose ac as shortest path.
from ac:

ac → e , f(e) = 4 + 17+5 = 26
ac → f , f(f) = 4 + 12+8 = 24 ✓

Choose ab as shortest path by backtracking.

ab → e, f(e) = 9 + 11+5 = 25

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Additional answer book No. 4110
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Choose acf as shortest.

acf → z , f(z) = 4+12+ 9 = 25 ✓
acf → d , f(d) = 4+12+14 +18 = 47

Choose acfz as shortest.

∴ The shortest path = acfz &
total path cost = 25.

7

Q6

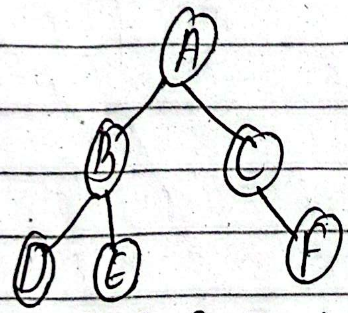
DFS - Depth First Search and BFS Breadth First Search are the two basic informed search.

DFS uses stack and BFS uses queue.

Algorithm for DFS:

1. initialize start node
2. visit the child node of start node before moving to neighbor nodes.
3. repeat the process until all child nodes of a node are not visited.
4. visiting child according first left child then right child.
5. After all child visited then go to neighbors. and repeat the process.
6. Stop.

example :



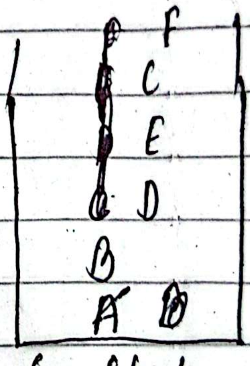


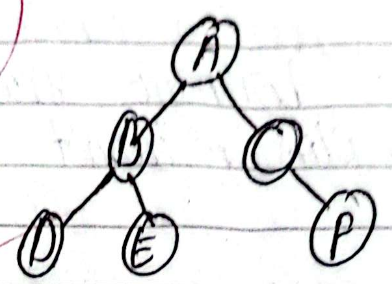
fig: Stack

DFS search: A-B-D-E-C-F.

BFS Algorithm:

1. initialize start node
2. visit the child node start node. Visit find all the immediate neighbors from left to right before moving children.
3. Then visit the child node of neighbors as step 2.
4. Repeat the process until all nodes are visited.
5. Stop

example: 6.5



A	B	C	D	E	F

fig: queue

BFS search: A - B - C - D - E - F.

As DFS uses stack it follows LIFO (Last in First out) and BFS uses queue follows FIFO (First in First Out) mechanisms.

2a Hill Climbing Algorithm

Hill climbining algorithm is a local search algorithm in which a component tries to climb hill.
For example N-queen problems are the Hill climbing problem. 4 Queen, 8 Queen etc.

There are 3 problems arises during Hill climbing:

a) Local maxima

It is the peak value ~~than~~ which is lower than global maxima. A climber is sured that it the destination but not.

b) Plateau

It the flatstate with no gratidient. As all direction same. Climber cann't decide where to go.

c) Ridges

The narrow space in hill. It cause to drowdown while need to uphill.

Simulated Annealing is introduced to avoids the problems and solve it.

Algorithm:

1. initialize start node
2. identify all nodes and neighbors.
3. Choose the best neighbor among the all neighbors.
4. Stop if no neighbor exist.

3.5

7b Informed and Uninformed search

Search is the method to find a desired node. There are two search techniques categorized as:

i) Informed Search

- Informed Search is the search in which we ~~are~~ ~~kn~~ know where is goal node.
- We choose the best path for that as we informed goal node destination.
- Greedy First Best Search, A* search are the informed search. These are faster and more effecient and effective than the uninformed search.
- It is mostly used in computer system.

ii) Uninformed Search 4

- Uninformed search is the search in which we don't know where is goal node.
- We have not knowledge which path is best.
- Depth First Search, Breadth First Search, Iterative Deeping search are the uninformed search.
- These search are costly and more time consuming process.